Name

Mathematics Advanced Year 12

Arithmetic Sequences and Series

Book 2

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Contents

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MAV-12-03 uses arithmetic and geometric sequences and series to model and solve problems

Arithmetic sequences and series

- Define a sequence a_n to be an arithmetic sequence, or arithmetic progression (AP), if every difference $a_n - a_{n-1}$ of successive terms is the same where $n \geq 2$, that is $a_n - a_{n-1} = d$ for some constant d called the common difference
- Develop the formula $a_n = a + (n-1)d$ for the nth term of an AP, where a is the first term and $n \geq 1$, and use it to solve problems
- Recognise that a_n is a linear function of n in an AP
- Develop the formula $S_n = \frac{n}{2}(a + a_n)$ for the nth partial sum of an AP, and use the formula to solve problems
- Develop the formula $S_n = \frac{n}{2}[2a + (n-1)d]$ for the nth partial sum of an AP, and use the formula to solve problems
- Apply the formulas for arithmetic sequences and their partial sums to model and solve growth and decay problems involving a quantity that is a linear function of time

Arithmetic Sequences

ⓘ Arithmetic Sequences

An **arithmetic sequence**, also called an **arithmetic progression (AP)** or **linear sequence**, is a sequence where there is a **common difference** between each pair of consecutive terms.

For example:

$$3, 7, 11, 15, 19, 23, \ldots$$

The common difference (d) is the same value that you add to each term to get the next term. It can be:

- Positive (sequence increases)
- Negative (sequence decreases)
- Zero (sequence stays constant)

We check whether a sequence is arithmetic by finding the differences between consecutive terms and seeing if they are constant.

$$a_n - a_{n-1} = d$$

ⓘ **Identify** whether a sequence is arithmetic.

✗	✓	✓	✗
1, 4, 9, 16, 25…	1, 4, 7, 10, 13…	55, 48, 41, 34, 27…	56, 48, 41, 34, 27…

Determine whether the following are arithmetic sequences.

a	46, 53, 40, 37…	**b**	1, 11, 111, 1111…	**c**	8, 27, 64, 125…
d	2, 4, 6, 8, 10, 12, 13…	**e**	$-2, -6, -10, -14\ldots$	**f**	2, 4, 8, 16, 32, …

ⓘ a and d

Let the first term of a sequence be a.
Let the common difference be d.

An arithmetic sequence is $a,\ a+d,\ a+2d,\ a+3d$ ...

Interpret a and d.

a Write the first five terms of an arithmetic sequence with $a = 130,\ d = 5$

b Write the first five terms of an arithmetic sequence with $a = 57, d = -2$

c Write the first five terms of an arithmetic sequence with $a = -82, d = 10$

ⓘ **Investigation** General term of any arithmetic sequence.

We can find the nth term formula of an arithmetic sequence in terms of a and d:

$a_1 = a$

$a_2 = a + d$

$a_3 = a + 2d$

$a_4 = a +$ .....

$a_5 =$ .....

...

$a_n =$ .....

ⓘ n^{th} term Formula for an Arithmetic Sequence

first term ↓ a, common difference ↓ d

$$a_n = a + (n-1)d$$

Example Finding and applying the rule of an arithmetic sequence.

Consider the arithmetic sequence: 5, 8, 11, 14 …

a Find the nth term formula for the sequence.

b What is the 12th term?

c What is the first term greater than 100?

a.

$d = 3, a = 5$

$a_n = a + (n-1)d$

$a_n = 5 + 3(n-1)$

$a_n = 2 + 3n$

b.

Substitute $n = 12$ into formula

$a_{12} = 2 + 3(12)$

$= 38$

c.

$3n + 2 > 100$

$3n > 98$

$n > 32.666\ldots$

∴ 33rd term is greater than 100.

$T_{33} = 3(33) + 2 = 101$

Practice

Consider the arithmetic sequence: 12, 14, 16, 18 …

a Find the nth term formula for the sequence.

b What is the 20th term?

c Which term of the sequence is 78?

$a_n = 10 + 2n$
$a_{20} = 50$
$78 = a_{34}$

Consider the arithmetic sequence: 17, 13, 9, 5 …

a Find the nth term formula for the sequence.

b What is the 15th term?

c What is the first term less than -50?

$a_n = 21 - 4n$
$a_{15} = -39$
$a_{18} = -51$

Example Further problems involving arithmetic sequences.

How many terms are there in the sequence 26, 35, 44, 53, … 917?

$a_n = a + (n-1)d$

$917 = 26 + (n-1)(9)$

$891 = 9(n-1)$

$99 = n - 1$

$\therefore n = 100$

There are 100 terms in the sequence.

Practice

How many terms are there in the sequence 17, 14, 11, 8, … –367?

129

Example Further problems involving arithmetic sequences.

The 5th term of an arithmetic sequence is 37 and the 8th term is 55.
Find the first term.

$a_5 = a + 4d = 37 \quad (1)$

$a_8 = a + 7d = 55 \quad (2)$

$(2) - (1)$:

$3d = 18$

$d = 6$

Substitute $d = 6$ into (1) to solve for a:

$a + 4(6) = 37$

$a = 13$

The first term is 13.

Practice

The 2nd term of an arithmetic sequence is 94 and the 10th term is 6.
Find the first term.

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ⓘ **Investigation** Linear relationships and arithmetic sequences

Consider the arithmetic sequence

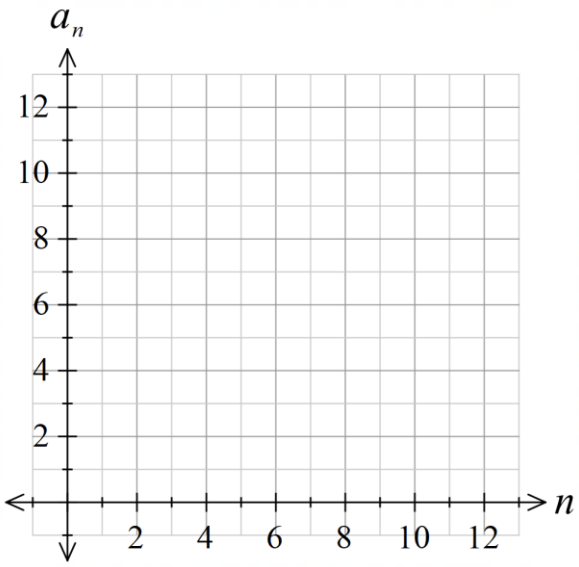
5, 7, 9, 11, 13, …

$a =$ …… $d =$ ……

So, the formula for the nth term is: $a_n =$ ……

Let's graph this sequence:

n	1	2	3	4	5
a_n	5	7	9	11	13



The sequence forms a …… relationship.

Using $y = mx + c$, write the equation for the line.

$a_n = \ldots n + \ldots$

How are your two equations,

$a_n = 5 + 2(n - 1)$ and $a_n = 2n + 3$ related?

How is the arithmetic sequence $a_n = 2n + 3$ different to the function $f(x) = 2x + 3$?
Hint: think about the domain!

1 Write out the next three terms of these sequences. They are all APs.

a 3, 8, 13, ...

18, 23, 28

b 35, 25, 15, ...

5, −5, −15

c 4½, 6, 7½, ...

9, 10½, 12

2 Write out the first four terms of the APs whose first terms and common differences are:

a $a = 3 \quad d = 2$

3, 5, 7, 9

b $a = 7 \quad d = -4$

7, 3, −1, −5

c $a = 30 \quad d = -11$

30, 19, 8, −3

d $a = -9 \quad d = 4$

−9, −5, −1, 3

e $a = 3\frac{1}{2} \quad d = -2$

3½, 1½, −½, −2½

f $a = 0.9 \quad d = 0.73$

0.9, 1.6, 2.3, 3

3 Find the differences $a_2 - a_1$ and $a_3 - a_2$ for each sequence to test whether it is an AP.
If the sequence is an AP, state the values of the first term a and the common difference d.

a $3, 7, 11, \ldots$

$a_2 - a_1 = 7 - 3 = 4$
$a_3 - a_2 = 11 - 7 = 4$
$AP: a = 3, d = 4$

b $11, 7, 3, \ldots$

$a_2 - a_1 = 7 - 11 = -4$
$a_3 - a_2 = 3 - 7 = -4$
$AP: a = 11, d = -4$

c $23, 34, 45, \ldots$

$a_2 - a_1 = 34 - 23 = 11$
$a_3 - a_2 = 45 - 34 = 11$
$AP: a = 23, d = 11$

d $-12, -7, -2, \ldots$

$a_2 - a_1 = -7 - (-12) = 5$
$a_3 - a_2 = -2 - (-7) = 5$
$AP: a = -12, d = 5$

e $-40, 20, -10, \ldots$

$a_2 - a_1 = 20 - (-40) = 60$
$a_3 - a_2 = -10 - 20 = -30$
not an AP

f $1, 11, 111, \ldots$

$a_2 - a_1 = 11 - 1 = 10$
$a_3 - a_2 = 111 - 11 = 100$
not an AP

g $8, -2, -12, \ldots$

$a_2 - a_1 = -2 - 8 = -10$
$a_3 - a_2 = -12 - (-2) = -10$
$AP: a = 8, d = -10$

h $-17, 0, 17, \ldots$

$a_2 - a_1 = 0 - (-17) = 17$
$a_3 - a_2 = 17 - 0 = 17$
$AP: a = -17, d = 17$

i $10, 7\frac{1}{2}, 5, \ldots 4$

$a_2 - a_1 = 7\frac{1}{2} - 10 = -2\frac{1}{2}$
$a_3 - a_2 = 5 - 7\frac{1}{2} = -2\frac{1}{2}$
$AP: a = 10, d = -2\frac{1}{2}$

4 Use the formula $a_n = a + (n-1)d$ to find the 11th term a_{11} of these APs:

a $a = 7 \quad d = 6$

AP where $a = 7, d = 6$
$a_{11} = 7 + 10(6)$
= 67

b $a = 15 \quad d = -7$

AP where $a = 15, d = -7$
$a_{11} = 15 + 10(-7)$
= -55

c $a = 10\frac{1}{2} \quad d = 4$

AP where $a = 10\frac{1}{2}, d = 4$
$a_{11} = 10\frac{1}{2} + 10(4)$
= 50½

5 Use the formula $a_n = a + (n-1)d$ to write an expression for the nth term a_n of these APs:

a $a = 1 \quad d = 4$

$a_n = a + (n-1)d$
$= 1 + (n-1)(4)$
$= 1 + 4n - 4$
$= 4n - 3$

b $a = 100 \quad d = -7$

$a_n = a + (n-1)d$
$= 100 + (n-1)(-7)$
$= 100 - 7n + 7$
$= 107 - 7n$

c $a = -13 \quad d = 6$

$a_n = a + (n-1)d$
$= -13 + (n-1)(6)$
$= -13 + 6n - 6$
$= 6n - 19$

6

a Find the first term a and the common difference d of the AP 6, 16, 26, ...

a = 6, d = 10

b Find the ninth term a_9, the 21st term a_{21} and the 100th term a_{100}.

$a_9 = 6 + 8(10)$

$= 86$

$a_{21} = 6 + 20(10)$

$= 206$

$a_{100} = 6 + 99(10)$

$= 996$

c Use the formula $a_n = a + (n - 1)d$ to find a formula for the nth term a_n.

$a_n = a + (n - 1)d$

$= 6 + (n - 1)(10)$

$= 6 + 10n - 10$

$= 10n - 4$

7 Find $a_3 - a_2$ and $a_2 - a_1$ to test whether each sequence is an AP.
If the sequence is an AP, use $a_n = a + (n - 1)d$ to find a formula for the nth term a_n.
Answer in simplified form.

a $8, 11, 14, \ldots$

$a_2 - a_1 = 11 - 8 = 3$
$a_3 - a_2 = 14 - 11 = 3$
$d = 3$
$a_n = a + (n - 1)d$
$= 8 + (n - 1)(3)$
$= 8 + 3n - 3$
$= 5 + 3n$

b $21, 15, 9, \ldots$

$a_2 - a_1 = 15 - 21 = -6$
$a_3 - a_2 = 9 - 15 = -6$
$d = -6$
$a_n = a + (n - 1)d$
$= 21 + (n - 1)(-6)$
$= 21 - 6n + 6$
$= 27 - 6n$

c $8, 4, 2, \ldots$

$a_2 - a_1 = 4 - 8 = -4$
$a_3 - a_2 = 2 - 4 = -2$
$not\ AP$

d $-3, 1, 5, \ldots$

$a_2 - a_1 = 1 - (-3) = 4$
$a_3 - a_2 = 5 - 1 = 4$
$d = 4$
$a_n = a + (n - 1)d$
$= -3 + (n - 1)(4)$
$= -3 + 4n - 4$
$= 4n - 7$

e $1¾, 3, 4¼, \ldots$

$a_2 - a_1 = 3 - 1¾ = \frac{5}{4}$
$a_3 - a_2 = 4¼ - 3 = \frac{5}{4}$
$d = \frac{5}{4}$
$a_n = a + (n - 1)d$
$= 1¾ + (n - 1)\left(\frac{5}{4}\right)$
$= \frac{7}{4} + \frac{5}{4}(n - 1)$
$= \frac{2 + 5n}{4}$

f $12, -5, -22, \ldots$

$a_2 - a_1 = -5 - 12 = -17$
$a_3 - a_2 = -22 - (-5) = -17$
$d = -17$
$a_n = a + (n - 1)d$
$= 12 + (n - 1)(-17)$
$= 12 - 17n + 17$
$= 29 - 17n$

$\sqrt{2}, 2\sqrt{2}, 3\sqrt{2}, \ldots$

$a_2 - a_1 = 2\sqrt{2} - \sqrt{2} = \sqrt{2}$
$a_3 - a_2 = 3\sqrt{2} - 2\sqrt{2} = \sqrt{2}$
$d = \sqrt{2}$
$a_n = a + (n - 1)d$
$= \sqrt{2} + (n - 1)\sqrt{2}$
$= n\sqrt{2}$

g $1, 4, 9, 16, \ldots$

$a_2 - a_1 = 4 - 1 = 3$
$a_3 - a_2 = 9 - 4 = 5$
not an AP

h $-2½, 1, 4½, \ldots$

$a^2 - a^1 = 1 - (-2½) = \frac{7}{2}$
$a^3 - a^2 = 4½ - 1 = \frac{7}{2}$
$d = \frac{7}{2}$
$a_n = a + (n - 1)d$
$= -2½ + (n - 1)\left(\frac{7}{2}\right)$
$= -\frac{5}{2} + \frac{7}{2}(n - 1)$
$= \frac{7n - 12}{2}$

8

a Use the formula $a_n = a + (n-1)d$ to find the nth term a_n of 165, 160, 155, ...

AP where $a = 165, d = -5$

$$\begin{aligned} a_n &= a + (n-1)d \\ &= 165 + (n-1)(-5) \\ &= 165 - 5n + 5 \\ &= 170 - 5n \end{aligned}$$

b Solve $a_n = 40$ to find the number of terms in the finite sequence 165, 160, 155, ..., 40.

$$\begin{aligned} a_n &= 40 \\ 170 - 5n &= 40 \\ 130 &= 5n \end{aligned}$$

$n = 26$ terms

c Solve $a_n < 0$ to find the first negative term of the sequence 165, 160, 155, ...9

$$\begin{aligned} a_n &< 0 \\ 170 - 5n &< 0 \\ 170 &< 5n \\ 34 &< n \end{aligned}$$

$\therefore a_{35} = -5$ is first negative term

9 Find a_n for each AP. Then solve $a_n < 0$ to find the first negative term.

a $20, 17, 14, \ldots$

AP where $a = 20, d = -3$

$$\begin{aligned} a_n &= a + (n-1)d \\ &= 20 + (n-1)(-3) \\ &= 23 - 3n \end{aligned}$$

$$\begin{aligned} a_n &< 0 \\ 23 - 3n &< 0 \\ 23 &< 3n \\ n &> 7\tfrac{2}{3} \end{aligned}$$

$\therefore a_8 = 23 - 3(8) = -1$

b $82, 79, 76, \ldots$

AP where $a = 82, d = -3$

$$\begin{aligned} a_n &= a + (n-1)d \\ &= 82 + (n-1)(-3) \\ &= 85 - 3n \end{aligned}$$

$$\begin{aligned} a_n &< 0 \\ 85 - 3n &< 0 \\ 85 &< 3n \\ n &> 28\tfrac{1}{3} \end{aligned}$$

$\therefore a_{29} = 85 - 3(29) = -2$

c $24\tfrac{1}{2}, 24, 23\tfrac{1}{2}, \ldots$

AP where $a = 24\tfrac{1}{2}, d = -\tfrac{1}{2}$

$$\begin{aligned} a_n &= a + (n-1)d \\ &= 24\tfrac{1}{2} + (n-1)(-\tfrac{1}{2}) \\ &= \left(\frac{49}{2}\right) - \left(\frac{1}{2}\right)(n-1) \\ &= \frac{50-n}{2} \end{aligned}$$

$$\begin{aligned} a_n &< 0 \\ \frac{50-n}{2} &< 0 \\ 50 - n &< 0 \\ 50 &< n \end{aligned}$$

$\therefore a_{51} = \dfrac{50-51}{2} = -\dfrac{1}{2}$

10 Find the number of terms in each finite sequence.

a $10, 12, 14, \ldots, 30$

$$\begin{aligned} a_n &= a + (n-1)d \\ 30 &= 10 + (n-1)(2) \\ 30 &= 10 + 2n - 2 \\ 22 &= 2n \end{aligned}$$

$n = 11$ terms

b $1, 4, 7, \ldots, 100$

$$\begin{aligned} a_n &= a + (n-1)d \\ 100 &= 1 + (n-1)(3) \\ 100 &= 1 + 3n - 3 \\ 102 &= 3n \end{aligned}$$

$n = 34$ terms

c $105, 100, 95, \ldots, 30$

$$\begin{aligned} a_n &= a + (n-1)d \\ 30 &= 105 + (n-1)(-5) \\ 30 &= 105 - 5n + 5 \\ 5n &= 80 \end{aligned}$$

$n = 16$ terms

d $100, 92, 84, \ldots, 4$

$$\begin{aligned} a_n &= a + (n-1)d \\ 4 &= 100 + (n-1)(-8) \\ 4 &= 100 - 8n + 8 \\ 8n &= 104 \end{aligned}$$

$n = 13$ terms

e $-12, -10\tfrac{1}{2}, -9, \ldots, 0$

$$\begin{aligned} a_n &= a + (n-1)d \\ 0 &= -12 + (n-1)(1\tfrac{1}{2}) \\ 12 &= \left(\frac{3}{2}\right)(n-1) \\ 8 &= n - 1 \end{aligned}$$

$n = 9$ terms

f $2, 5, 8, \ldots, 200011$

$$\begin{aligned} a_n &= a + (n-1)d \\ 2000 &= 2 + (n-1)(3) \\ 1998 &= 3n - 3 \\ 2001 &= 3n \end{aligned}$$

$n = 667$ terms

11 The nth term of an arithmetic sequence is $a_n = 7 + 4n$.

a Write out the first four terms, and hence find the values of a and d.

$11, 15, 19, 23$

$a = 11, d = 4$

b Which term of the sequence is 815?

$a_n = 815$

$7 + 4n = 815$

$4n = 808$

$n = 202$

202nd term

c Find the last term less than 1000 and the first term greater than 1000.

$a_n < 1000$

$7 + 4n < 1000$

$4n < 993$

$n < 248.25$

$\therefore a_{248} = 7 + 4(248) = 999$ is last term less than 1000

$\therefore a_{249} = 7 + 4(249) = 1003$ is first term greater than 1000

d Find how many terms there are between 200 and 300.

$200 < a_n < 300$

$200 < 7 + 4n < 300$

$193 < 4n < 293$

$48.25 < n < 73.25$

$\therefore a_{49}, \ldots, a_{73}$ lie between 200 and 300

Number of terms = 73 - 49 + 1 = 25 terms

12

a Let a_n be the sequence 8, 16, 24, ... of positive multiples of 8.

i Show that the sequence is an AP, and find a formula for a_n.

$a_2 - a_1 = 16 - 8 = 8$

$a_3 - a_2 = 24 - 16 = 8$

$d = 8$, so it is an AP

AP where $a = 8, d = 8$

$a_n = a + (n - 1)d$

$= 8 + (n - 1)(8)$

$= 8n$

ii Find the first term of the sequence greater than 500 and the last term less than 850.

$a_n > 500$

$8n > 500$

$n > 62.5$

$\therefore a_{63} = 8(63) = 504$ is first term greater than 500

iii Hence find the number of multiples of 8 between 500 and 850.

$a_n < 850$

$8n < 850$

$n < 106.25$

$\therefore a_{106} = 8(106) = 848$ is last term less than 850

Number of terms $= 106 - 63 + 1 = 44$ terms

13 Find the number of multiples of 11 between 1000 and 2000.

$a_n = 11n$

$1000 < 11n < 2000$

$90.9\ldots < n < 181.8\ldots$

Number of multiples of 11 between 1000 and 2000 $= 181 - 91 + 1 = 91$ terms

14

a The first term of an AP is $a = 7$ and the fourth term is $a_4 = 16$.
Use the formula $a_n = a + (n-1)d$ to find the common difference d.
Then write down the first four terms.

AP where $a = 7, a_4 = 16$

$a_4 = a + (4-1)d$

$16 = 7 + 3d$

$9 = 3d$

$d = 3$

First four terms: 7, 10, 13, 16

b Find the 20th term of an AP with first term 28 and 11th term 108.

AP where $a = 28, a_{11} = 108$

$a_{11} = a + (11-1)d$

$108 = 28 + 10d$

$80 = 10d$

$d = 8$

$a_{20} = a + (20-1)d$

$= 28 + 19(8)$

$= 180$

c Find the 100th term of an AP with first term 32 and 20th term -6.

AP where $a = 32, a_{20} = -6$

$a_{20} = a + (20-1)d$

$-6 = 32 + 19d$

$-38 = 19d$

$d = -2$

$a_{100} = a + (100-1)d$

$= 32 + 99(-2)$

$= -166$

15 A windows shop charges \$500 for the first window, then \$300 for each additional window.

a Write down the cost of 1 window, 2 windows, 3 windows, 4 windows, ...

\$500, \$800, \$1100, \$1400

b Use the formula $a_n = a + (n-1)d$ to find the cost of 15 windows.

AP where $a = 500, d = 300$

$a_{15} = a + (15-1)d$

$= 500 + 14(300)$

$= \$4700$

c Find the cost of n windows.

$a_n = a + (n-1)d$

$= 500 + (n-1)(300)$

$= 500 + 300n - 300$

$= 200 + 300n$

d Find the maximum number of windows whose total cost is less than \$10 000.

$a_n < 10000$

$200 + 300n < 10000$

$300n < 9800$

$n < 32\frac{2}{3}$

$\therefore$ maximum number $= 32$ *windows*

16 A principal of \$2000 is invested at 6% per annum simple interest.
Let $\$A_n$ be the total amount (principal plus interest) at the end of n years.

a Write out the values of A_1, A_2, A_3 and A_4.

Interest per year = 6% of \$2000 = \$120

$A_1 = \$2120, A_2 = \$2240, A_3 = \$2360, A_4 = \$2480$

b Use the formula $a_n = a + (n-1)d$ to find a formula for A_n, and evaluate A_{12}.

AP where $a = 2120, d = 120$

$A_n = a + (n-1)d$

$= 2120 + (n-1)(120)$

$= 2120 + 120n - 120$

$= 2000 + 120n$

$A_{12} = 2000 + 120(12)$

$= \$3440$

c How many years will it take before the total amount exceeds \$6000?

$A_n > 6000$

$2000 + 120n > 6000$

$120n > 4000$

$n > 33\frac{1}{3}$

$\therefore$ 34 years

17

a Write down the first few terms of the AP generated by substituting the positive integers into the linear function $f(x) = 12 - 3x$. Then write down a formula for the nth term.

$f(x) = 12 - 3x$

$f(1) = 12 - 3(1) = 9$

$f(2) = 12 - 3(2) = 6$

$f(3) = 12 - 3(3) = 3$

$f(4) = 12 - 3(4) = 0$

$f(5) = 12 - 3(5) = -3$

$a_n = 12 - 3n$

b

i Find the formula of the nth term a_n of the AP $-3, -1, 1, 3, 5, \ldots$ Then write down the linear function $f(x)$ that generates this AP when the positive integers are substituted into it.

$a = -3, d = 2$

$a_n = a + (n-1)d$

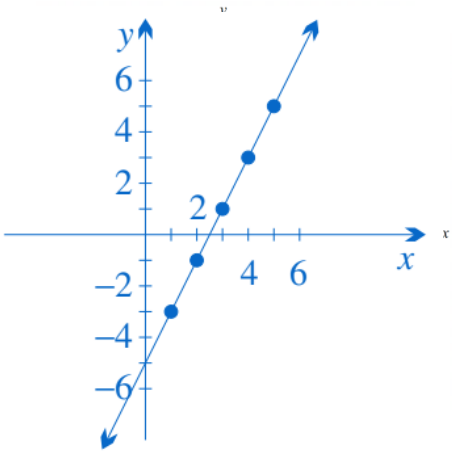
$= -3 + (n-1)(2)$

$= -3 + 2n - 2$

$= 2n - 5$

$f(x) = 2x - 5$

ii Graph the function and mark the points $(1, -3), (2, -1), (3, 1), (4, 3), (5, 5)$. How is the arithmetic sequence similar and different to the function?



The points of the sequence lie on the line.

However, the domain of the sequence is not continuous, since n is restricted to positive integers.

18 Find the common difference of each AP. Then find x if $a_{11} = 36$.

a $5x - 9, 5x - 5, 5x - 1, \ldots$

$a_{11} = 36$

$a + (11-1)d = 36$

$(5x - 9) + 10(4) = 36$

$5x - 9 + 40 = 36$

$5x + 31 = 36$

$5x = 5$

$x = 1$

b $16, 16 + 6x, 16 + 12x, \ldots 19$

$a_{11} = 36$

$a + (11-1)d = 36$

$16 + 10(6x) = 36$

$16 + 60x = 36$

$60x = 20$

$x = \frac{1}{3}$

19 Find the common difference of each AP. Then find a formula for the nth term a_n.

a $\log_3 2,\ \log_3 4,\ \log_3 8, \ldots$

$$d = log^3 4 - log^3 2 = \log_3\left(\frac{4}{2}\right) = log_3 2$$
$$a_n = a + (n-1)d$$
$$= \log_3 2 + (n-1)\log_3 2$$
$$= n\log_3 2$$

b $\log_a 54,\ \log_a 18,\ \log_a 6, \ldots$

$$d = \log_a 18 - \log_a 54 = \log_a\left(\frac{18}{54}\right) = \log_a\left(\frac{1}{3}\right) = -\log_a 3$$
$$a_n = a + (n-1)d$$
$$= \log_a 54 + (n-1)(-\log_a 3)$$
$$= \log_a 54 - (n-1)\log_a 3$$
$$= \log_a(2 \times 3^3) + (4-n)\log_a 3$$
$$= \log_a 2 + 3\log_a 3 + (4-n)\log_a 3$$
$$= \log_a 2 + (4-n)\log_a 3$$

c $x - 3y, 2x + y, 3x + 5y, \ldots$

$$d = (2x + y) - (x - 3y) = x + 4y$$
$$a_n = a + (n-1)d$$
$$= (x - 3y) + (n-1)(x + 4y)$$
$$= x - 3y + nx + 4ny - x - 4y$$
$$= nx + (4n - 7)y$$

d $5 - 6\sqrt{5}, 1 + \sqrt{5}, -3 + 8\sqrt{5}, \ldots$

$$d = (1 + \sqrt{5}) - (5 - 6\sqrt{5}) = -4 + 7\sqrt{5}$$
$$a_n = a + (n-1)d$$
$$= (5 - 6\sqrt{5}) + (n-1)(-4 + 7\sqrt{5})$$
$$= 5 - 6\sqrt{5} - 4n + 4 + 7n\sqrt{5} - 7\sqrt{5}$$
$$= 9 - 4n + (7n - 13)\sqrt{5}$$

e $1.36, -0.52, -2.4, \ldots$

$$d = -0.52 - 1.36 = -1.88$$
$$a_n = a + (n-1)d$$
$$= 1.36 + (n-1)(-1.88)$$
$$= 1.36 - 1.88n + 1.88$$
$$= 3.24 - 1.88n$$

f $\log_a 3x^2,\ \log_a 3x,\ \log_a 3, \ldots$

$$d = \log_a 3x - \log_a 3x^2 = \log_a\left(\frac{3x}{3x^2}\right) = \log_a\left(\frac{1}{x}\right) = -\log_a x$$
$$a_n = a + (n-1)d$$
$$= \log_a 3x^2 + (n-1)(-\log_a x)$$
$$= \log_a 3 + 2\log_a x - (n-1)\log_a x$$
$$= \log_a 3 + (3-n)\log_a x$$

Partial Sum of an Arithmetic Series

Investigation Partial sum of a Geometric Series.

Reminder that the partial sum of a sequence is:

$$S_n = a_1 + a_2 + a_3 + \cdots a_n$$

Consider the partial sum of the arithmetic sequence below:

$$S_6 = 5 + 15 + 25 + 35 + 45 + 55$$

A quick way of adding up everything is by **reversing the sum**, then pairing up the terms.

$$S_6 = 55 + 45 + 35 + 25 + 15 + 5$$

$$2S_6 = \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

$$\therefore S_6 = \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

Use the above method to evaluate:

$$S_8 = 9 + 14 + 19 + 24 + 29 + 34 + 39 + 44$$

We will generalise to form an equation; we will first define let the nth term be a_n.

$$S_n = a + (a + d) + (a + 2d) + (a + 3d) + \cdots a_n$$

We can write an alternate form of the formula by substituting $a_n = a + (n - 1)d$

The Partial Sum of an Arithmetic Series

$$S_n = \frac{n}{2}(a + a_n) \qquad S_n = \frac{n}{2}[2a + (n-1)d]$$

⚠ Usually n is not given, you need to find it using $a_n = a + (n-1)d$

Example Finding the partial sum of an arithmetic series.

Consider the arithmetic series: $100 + 94 + 88 + 82 \ldots$
Find S_{10}

$a = 100, d = -6$

$$S_{10} = \frac{10}{2}\big(2(100) + (10-1)(-6)\big)$$
$$= 730$$

Practice

Consider the arithmetic series: $56 + 51 + 46 + \cdots$
Find S_{21}

126

Consider the arithmetic series: $41 + 45 + 49 + \cdots + 401$.
Find the number of terms in the partial sum and hence evaluate the partial sum of the series.

20 111

Example Further problems involving sums of arithmetic sequences.

Find the least value of n for which the partial sum S_n of $40 + 37 + 34 + \cdots$ is negative.

$a = 40, d = -3$

Let $S_n < 0$

$\frac{n}{2}\big(2(40) + (n-1)(-3)\big) < 0$

$\frac{n}{2}(80 - 3(n-1)) < 0$

$\frac{n}{2}(83 - 3n) < 0$

$83 - 3n < 0$ Note: we can safely divide by $\frac{n}{2}$ as n must be positive.

$3n > 83$

$n > 27.66..$ The 28th partial sum is negative.

Practice

Find the value of n for which the partial sum S_n of $2 + 11 + 20 + \cdots$ is equal to 618.

12

How many terms of the series $4 + 7 + 10 + \cdots$ are needed to give a sum of 531?

18

Example Further problems involving sums of arithmetic sequences.

The sum of the first 10 terms of an arithmetic sequence is 0.
The sum of the first and second terms is 24.
Find the sum of the first three terms.

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$S_{10} = 0$$

$$\frac{10}{2}(2a + (10-1)d) = 0$$

$$5(2a + 9d) = 0$$

$$10a + 45d = 0 \quad (1)$$

$$S_2 = 24$$

$$\frac{2}{2}(2a + (2-1)d) = 24$$

$$2a + d = 24 \quad (2)$$

$$(2) \times 5: \ 10a + 5d = 120 \quad (3)$$

$(1) - (3)$:

$$40d = -120$$

$$d = -3$$

Substitute d into (2) and solve for a:

$$a = 13.5$$

$$S_3 = \frac{3}{2}\big(2(13.5) + (3-1)(-3)\big)$$

$$= 31.5$$

Practice

The 6th term of an arithmetic sequence is 23 and the sum of the first 10 terms is 210.
Find the sum of the first 20 terms of the sequence.

820

The sum of the first four terms of an arithmetic series is 34 and the sum of the next four terms is 146. Find the sum of the ninth and tenth terms

115

1 Use $S_n = \frac{n}{2}(2a + (n-1)d)$ to find the sum S_6 of the first 6 terms of the series with:

a $a = 5 \quad d = 10$

$S^6 = \frac{n}{2}(2a + (n-1)d)$
$= \left(\frac{6}{2}\right)(2(5) + 5(10))$
$= 180$

b $a = 8 \quad d = 2$

$S_6 = \left(\frac{6}{2}\right)(2(8) + 5(2))$
$= 78$

c $a = -3 \quad d = -9$

$S_6 = \left(\frac{6}{2}\right)(2(-3) + 5(-9))$
$= -153$

d $a = -7 \quad d = -124$

$S_6 = \left(\frac{6}{2}\right)(2(-7) + 5(-12))$
$= -222$

2 Use the formula $S_n = \frac{n}{2}(2a + (n-1)d)$ to find the sum of the stated number of terms.

a 2 + 5 + 8 + ... (12 terms)

$S_{12} = \left(\frac{12}{2}\right)(2(2) + 11(3))$
$= 222$

b 40 + 33 + 26 + ... (21 terms)

$S_{21} = \left(\frac{21}{2}\right)(2(40) + 20(-7))$
$= -630$

c -6 - 2 + 2 + ... (200 terms)

$S_{200} = \left(\frac{200}{2}\right)(2(-6) + 199(4))$
$= 78400$

d 33 + 30 + 27 + ... (23 terms)

$S_{23} = \left(\frac{23}{2}\right)(2(33) + 22(-3))$
$= 0$

e -10 - 7½ - 5 + ... (13 terms)

$S_{13} = \left(\frac{13}{2}\right)(2(-10) + 12(2½))$
$= 65$

f 10½ + 10 + 9½ + ... (40 terms)

$S_{40} = \left(\frac{40}{2}\right)(2(10½) + 39(-½))$
$= 30$

3 First use the formula $a_n = a + (n-1)d$ to find how many terms there are in each sum.
Then use the formula $S_n = \frac{n}{2}(a + \ell)$ to find the sum, where ℓ is the last term a_n.

a 50 + 51 + 52 + ... + 150

$a_n = a + (n-1)d$
$150 = 50 + (n-1)(1)$
$100 = n - 1$
$n = 101$
$S_n = \frac{n}{2}(a + \ell)$
$= \left(\frac{101}{2}\right)(50 + 150)$
$= 10100$

b 8 + 15 + 22 + ... + 92

$a_n = a + (n-1)d$
$92 = 8 + (n-1)(7)$
$84 = 7n - 7$
$91 = 7n$
$n = 13$
$S_n = \left(\frac{13}{2}\right)(8 + 92)$
$= 650$

c -10 - 3 + 4 + ... + 60

$a_n = a + (n-1)d$
$60 = -10 + (n-1)(7)$
$70 = 7n - 7$
$77 = 7n$
$n = 11$
$S_n = \left(\frac{11}{2}\right)(-10 + 60)$
$= 275$

d 4 + 7 + 10 + ... + 301

$a_n = a + (n-1)d$
$301 = 4 + (n-1)(3)$
$297 = 3n - 3$
$300 = 3n$
$n = 100$
$S_n = \left(\frac{100}{2}\right)(4 + 301)$
$= 15250$

e 6½ + 11 + 15½ + ... + 51½

$a_n = a + (n-1)d$
$51\frac{1}{2} = 6\frac{1}{2} + (n-1)(4\frac{1}{2})$
$45 = \left(\frac{9}{2}\right)(n-1)$
$10 = n - 1$
$n = 11$
$S_n = \left(\frac{11}{2}\right)(6\frac{1}{2} + 51\frac{1}{2})$
$= 319$

f -1⅓ + ⅓ + 2 + ... + 13⅔

$a_n = a + (n-1)d$
$\frac{41}{3} = -\frac{4}{3} + (n-1)\left(\frac{5}{3}\right)$
$\frac{45}{3} = \left(\frac{5}{3}\right)(n-1)$
$9 = n - 1$
$n = 10$
$S_n = \left(\frac{10}{2}\right)\left(-\frac{4}{3} + \frac{41}{3}\right)$
$= 61\frac{2}{3}$

4 Find these sums by any appropriate method.

a 2 + 4 + 6 + ... + 1000

$a = 2, d = 2, \ell = 1000$
$a_n = a + (n-1)d$
$1000 = 2 + (n-1)(2)$
$998 = 2n - 2$
$1000 = 2n$
$n = 500$
$S_n = \left(\frac{500}{2}\right)(2 + 1000)$
$= 250500$

b 1000 + 1001 + ... + 3000

$a = 1000, d = 1, \ell = 3000$
$a_n = a + (n-1)d$
$3000 = 1000 + (n-1)(1)$
$2000 = n - 1$
$n = 2001$
$S_n = \left(\frac{2001}{2}\right)(1000 + 3000)$
$= 4002000$

c 1 + 5 + 9 + ... (40 terms)

$S_n = \left(\frac{40}{1}\right)(2(1) + 39(4))$
$= 20(2 + 156)$
$= 3160$

d 10 + 30 + 50 + ... (12 terms)

$S_n = \left(\frac{12}{2}\right)(2(10) + 11(20))$
$= 6(20 + 220)$
$= 1440$

5 Find simplify the expression for the sum of the first n terms of each series.

a $5 + 10 + 15 + \ldots$

$a = 5, d = 5$

$S_n = \frac{n}{2}(2a + (n-1)d)$

$= \frac{n(10 + 5n - 5)}{2}$

$= \frac{n(5 + 5n)}{2}$

$= \frac{5n(1 + n)}{2}$

b $10 + 13 + 16 + \ldots$

$a = 10, d = 3$

$S_n = \frac{n(20 + 3n - 3)}{2}$

$= \frac{n(17 + 3n)}{2}$

c $3 + 7 + 11 + \ldots$

$a = 3, d = 4$

$S_n = \frac{n}{2}(6 + 4n - 4)$

$= \frac{n(2 + 4n)}{2}$

$= n(1 + 2n)$

d $-9 - 4 + 1 + \ldots$

$a = -9, d = 5$

$S_n = \frac{n}{2}(2(-9) + (n-1)(5))$

$= \frac{n(-18 + 5n - 5)}{2}$

$= \frac{n(5n - 23)}{2}$

e $5 + 4\frac{1}{2} + 4 + \ldots$

$a = 5, d = -\frac{1}{2}$

$S_n = \frac{n}{2}(2(5) + (n-1)(-\frac{1}{2}))$

$= \frac{n}{2}\left(10 - \left(\frac{1}{2}\right)n + \left(\frac{1}{2}\right)\right)$

$= \frac{n}{2}\left(\frac{21}{2} - \frac{n}{2}\right)$

$= \frac{n(21 - n)}{4}$

f $(1 - \sqrt{2}) + 1 + (1 + \sqrt{2}) + \cdots$

$a = 1 - \sqrt{2}, d = \sqrt{2}$

$S_n = \frac{n}{2}(2(1 - \sqrt{2}) + (n-1)\sqrt{2})$

$= \frac{n(2 - 2\sqrt{2} + n\sqrt{2} - \sqrt{2})}{2}$

$= \frac{n(2 + n\sqrt{2} - 3\sqrt{2})}{2}$

6 An advertising graduate earns \$28000 per annum in her first year, then each successive year her salary rises by \$1600. What are her total earnings over 10 years?

AP where $a = 28000, d = 1600, n = 10$

$S_{10} = \left(\frac{10}{2}\right)(2(28000) + 9(1600))$

$= \$352000$

7 By substituting appropriate values of k, find the first term a and last term ℓ of each sum. Then evaluate the sum using $S_n = \frac{n}{2}(a + \ell)$. (Note that all four series are APs.

a $\sum_{k=1}^{200}(600 - 2k)$

$a = 600 - 2(1) = 598$

$\ell = 600 - 2(200) = 200$

$n = 200$

$S_n = \left(\frac{200}{2}\right)(598 + 200)$

$= 79800$

b $\sum_{k=1}^{61}(93 - 3k)$

$a = 93 - 3(1) = 90$

$\ell = 93 - 3(61) = -90$

$n = 61$

$S_n = \left(\frac{61}{2}\right)(90 - 90)$

$= 0$

c $\sum_{k=1}^{40}(3k - 50)$

$a = 3(1) - 50 = -47$

$\ell = 3(40) - 50 = 70$

$n = 40$

$S_n = \left(\frac{40}{2}\right)(-47 + 70)$

$= 460$

d $\sum_{k=10}^{30}(5k + 3)$

$a = 5(10) + 3 = 53$

$\ell = 5(30) + 3 = 153$

$n = 30 - 10 + 1 = 21$

$S_n = \left(\frac{21}{2}\right)(53 + 153)$

$= 2163$

8 Find the sums of these arithmetic sequences, whose terms are logarithms.

a $\log_a 2 + \log_a 4 + \log_a 8 + \ldots + \log_a 1024$

$\log_a 1024 = \log_a 2^{10} = 10\log_a 2$

$a = \log_a 2, d = \log_a 2, \ell = 10\log_a 2$

$a_n = a + (n-1)d$

$10\log_a 2 = \log_a 2 + (n-1)\log_a 2$

$10 = 1 + n - 1$

$n = 10$

$S_n = \frac{10}{2}(\log_a 2 + 10\log_a 2)$

$= 5(11 \log_a\ 2)$

$= 55\log_a 2$

b $\log_5 243 + \log_5 81 + \log_5 27 + \ldots + \log_5\left(\frac{1}{243}\right)$

$243 = 3^5, 81 = 3^4, 27 = 3^3, \frac{1}{243} = 3^{-5}$

$a = 5\log_5 3, d = -\log_5 3, \ell = -5\log_5 3$

$a_n = a + (n-1)d$

$-5\log_5 3 = 5\log_5 3 + (n-1)(-\log_5 3)$

$-5 = 5 - n + 1$

$-5 = 6 - n$

$n = 11$

$S_n = \left(\frac{11}{2}\right)(5\log_5 3 - 5\log_5 3)$

$= 0$

c $\log_b 36 + \log_b 18 + \log_b 9 + \ldots + \log_b\left(\frac{9}{8}\right)$

$a = \log_b 36 = 2\log_b 2 + 2\log_b 3$

$d = \log_b 18 - \log_b 36 = \log_b\left(\frac{18}{36}\right) = \log_b\left(\frac{1}{2}\right) = -\log_b 2$

$\ell = \log_b\left(\frac{9}{8}\right) = 2\log_b 3 - 3\log_b 2$

$a_n = a + (n-1)d$

$2\log_b 3 - 3\log_b 2 = 2\log_b 2 + 2\log_b 3 + (n-1)(-\log_b 2)$

$-3 \log_b\ 2 = 2 \log_b\ 2 - n \log_b\ 2 + \log_b\ 2$

$-3 = 3 - n$

$n = 6$

$S_n = \left(\frac{1}{2}\right)(6)\left(\log_b 36 + \log_b\left(\frac{9}{8}\right)\right)$

$= 3\left(\log_b 36 + \log_b\left(\frac{9}{8}\right)\right)$

$= 3\log_b\left(36 \times \frac{9}{8}\right)$

$= 3\log_b\left(\frac{81}{2}\right)$

$= 3(4\log_b 3 - \log_b 2)$

9 Solve these questions using the formula $S_n = \frac{n}{2}(a + \ell)$ whenever possible, otherwise use the formula $S_n = \frac{n}{2}(2a + (n - 1)d)$.

a Find the last term if a series with 10 terms and first term -23 has sum -5.

$$S_n = \frac{n}{2}(a + \ell)$$
$$-5 = \left(\frac{10}{2}\right)(-23 + \ell)$$
$$-5 = 5(-23 + \ell)$$
$$-1 = -23 + \ell$$
$$\ell = 22$$

b Find the first term if a series with 40 terms and last term $8½$ has sum 28.

$$S_n = \frac{n}{2}(a + \ell)$$
$$28 = \left(\frac{40}{2}\right)(a + 8½)$$
$$28 = 20(a + 8½)$$
$$1.4 = a + 8.5$$
$$a = -7.1$$

c Find the common difference if a series with 8 terms and first term 5 has sum 348.

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$
$$348 = \left(\frac{8}{2}\right)(2(5) + 7d)$$
$$348 = 4(10 + 7d)$$
$$87 = 10 + 7d$$
$$77 = 7d$$
$$d = 11$$

d Find the first term if a series with 15 terms and difference $\frac{2}{7}$ has sum -15.

$$S_n = \left(\frac{n}{2}\right)(2a + (n - 1)d)$$
$$-15 = \left(\frac{15}{2}\right)(2a + 14(2/7))$$
$$-15 = \left(\frac{15}{2}\right)(2a + 4)$$
$$-2 = 2a + 4$$
$$-6 = 2a$$
$$a = -3$$

10

a Show that the sum to n terms of the AP $60 + 52 + 44 + 36 + \ldots$ is $S_n = 4n(16 - n)$.

$a = 60, d = -8$

$S_n = \frac{n}{2}(2a + (n-1)d)$

$= \frac{n}{2}(2(60) + (n-1)(-8))$

$= \frac{n}{2}(120 - 8n + 8)$

$= \frac{n}{2}(128 - 8n)$

$= 4n(16 - n)$ as required

b Hence find how many terms must be taken to make the sum:

i Zero

$4n(16 - n) = 0$

$n = 0$ or $n = 16$

$\therefore$ 16 terms ($n > 0$)

ii Negative

$4n(16 - n) < 0$

$n(16 - n) < 0$

$n > 16$

$\therefore$ more than 16 terms

c Find the two values of n for which the sum S_n is 220.

$4n(16 - n) = 220$

$64n - 4n^2 = 220$

$4n^2 - 64n + 220 = 0$

$n^2 - 16n + 55 = 0$

$(n - 5)(n - 11) = 0$

$n = 5$ or $n = 11$

d Show that $S_n = -144$ has two integer solutions, but that only one has meaning.

$S_n = -144$

$4n(16 - n) = -144$

$64n - 4n^2 = -144$

$4n^2 - 64n - 144 = 0$

$n^2 - 16n - 36 = 0$

$(n - 18)(n + 2) = 0$

$n = 18$ or $n = -2$

Only $n = 18$ has meaning (n must be positive)

e For what values of n does the sum S_n exceed 156?

$4n(16 - n) > 156$

$64n - 4n^2 > 156$

$4n^2 - 64n + 156 < 0$

$n^2 - 16n + 39 < 0$

$(n - 3)(n - 13) < 0$

By sketching the parabola,

$\therefore 3 < n < 13$

$\therefore n = 4, 5, 6, \ldots, 12$

f Prove that no sum S_n can exceed 256.

$S_n = 4n(16 - n) = 64n - 4n^2$

Completing square: $-4(n^2 - 16n) = -4(n - 8)^2 + 256$

Consider the parabola, it is concave down with maximum turning point at 256 when $n = 8$

so, no sum can exceed 256

11 First use the formula $S_n = \frac{n}{2}(2a + (n-1)d)$ to find the sum S_n for each arithmetic series. Then use quadratic equations to find the number of terms if S_n has the given value.

a $42 + 40 + 38 + \ldots$ where $S_n = 0$

$a = 42, d = -2$
$S_n = \frac{n}{2}(2(42) + (n-1)(-2))$
$= \frac{n}{2}(84 - 2n + 2)$
$= \frac{n}{2}(86 - 2n)$
$= n(43 - n)$

$S_n = 0$
$n(43 - n) = 0$
$n = 43\ (n > 0)$
$\therefore$ 43 terms

b $60 + 57 + 54 + \ldots$ where $S_n = 0$

$a = 60, d = -3$
$S_n = \frac{n}{2}(2(60) + (n-1)(-3))$
$= \frac{n}{2}(120 - 3n + 3)$
$= \frac{n}{2}(123 - 3n)$
$= \frac{3n}{2}(41 - n)$

$S_n = 0$
$\frac{3n}{2}(41 - n) = 0$
$n = 41\ (n > 0)$
$\therefore$ 41 terms

c $45 + 51 + 57 + \ldots$ where $S_n = 153$

$a = 45, d = 6$
$S_n = \frac{n}{2}(2(45) + (n-1)(6))$
$= \frac{n}{2}(90 + 6n - 6)$
$= \frac{n}{2}(84 + 6n)$
$= 3n(14 + n)$

$S_n = 153$
$3n(14 + n) = 153$
$3n^2 + 42n = 153$
$n^2 + 14n - 51 = 0$
$(n-3)(n+17) = 0$
$n = 3\ (n > 0)$
$\therefore$ 3 terms

d $2\frac{1}{2} + 3 + 3\frac{1}{2} + \ldots$ where $S_n = 22\frac{1}{2}$

$a = \frac{5}{2}, d = \frac{1}{2}$
$S_n = \frac{n}{2}\left(2\left(\frac{5}{2}\right) + (n-1)\left(\frac{1}{2}\right)\right)$
$= \frac{n}{2}\left(5 + \frac{n}{2} - \frac{1}{2}\right)$
$= \frac{n}{2}\left(\frac{9}{2} + \frac{n}{2}\right)$
$= \frac{n}{4}(9 + n)$

$S_n = 22\frac{1}{2}$
$\frac{n}{4}(9 + n) = \frac{45}{2}$
$n(9 + n) = 90$
$n^2 + 9n - 90 = 0$
$(n-6)(n+15) = 0$
$n = 6\ (n > 0)$
$\therefore$ 6 terms

12 Find the first term and the number of terms if a series has:

a $d = 4, \ell = 32$ and $S_n = 0$

$S_n = \frac{n}{2}(a + \ell)$
$0 = \frac{n}{2}(a + 32)$
$a = -32$

$a_n = a + (n-1)d$
$32 = -32 + (n-1)(4)$
$64 = 4n - 4$
$68 = 4n$
$n = 17$

b $d = -3, \ell = -10$ and $S_n = 55$

$S_n = \frac{n}{2}(a + \ell)$
$55 = \frac{n}{2}(a - 10)$
$110 = n(a - 10)$ ... (1)

$a_n = a + (n-1)d$
$-10 = a + (n-1)(-3)$
$-10 = a - 3n + 3$
$a = 3n - 13$ ... (2)

Substitute (2) into (1):
$110 = n(3n - 13 - 10)$
$110 = n(3n - 23)$
$(3n+10)(n-11) = 0$
$n = 11\ (n > 0)$

$a = 3(11) - 13 = 20$

13

a Logs of wood are stacked with 10 on the top row, 11 on the next, and so on. If there are 390 logs, find the number of rows, and the number of logs on the bottom row.

$AP\ where\ a = 10, d = 1, S_n = 390$

$S_n = \frac{n}{2}(2a + (n-1)d)$

$390 = \frac{n}{2}(2(10) + (n-1)(1))$

$390 = \frac{n}{2}(20 + n - 1)$

$780 = n(19 + n)$

$n^2 + 19n - 780 = 0$

$(n - 20)(n + 39)$

$n = 20\ (n > 0)$

Bottom row: $a_n = a + (n-1)d = 10 + 19(1) = 29$ logs

b A stone dropped from the top of a 245-metre cliff falls 5 metres in the first second, 15 metres in the second second, and so on in arithmetic sequence. Find a formula for the distance after n seconds, and find how long the stone takes to fall to the ground.

$AP: 5, 15, 25, \ldots$

$a = 5, d = 10$

Distance after n seconds:

$S_n = \frac{n}{2}(2(5) + (n-1)(10))$

$= \frac{n}{2}(10 + 10n - 10)$

$= 5n^2$

$S_n = 245$

$5n^2 = 245$

$n^2 = 49$

$n = 7$ seconds

14 The sum of the first 10 terms of an AP is zero, and the 10th term is -9. Find the first and second terms.

$S_{10} = 0, a_{10} = -9$

$S_{10} = \left(\frac{10}{2}\right)(a_1 + a_{10})$

$0 = 5(a_1 - 9)$

$a_1 = 9$

$a_{10} = a_1 + 9d$

$-9 = 9 + 9d$

$-18 = 9d$

$d = -2$

$a_2 = a_1 + d = 9 - 2 = 7$

15 The sum of the 1st and 4th terms of an AP is 16, and the sum of the third and eighth terms is 4. Find the sum of the first 10 terms.

$a_1 + a_4 = 16 \ldots (1)$

$a_3 + a_8 = 4 \ldots (2)$

$a_1 + (a_1 + 3d) = 16$

$2a_1 + 3d = 16 \ldots (3)$

$(a_1 + 2d) + (a_1 + 7d) = 4$

$2a_1 + 9d = 4 \ldots (4)$

Subtract (3) from (4):

$6d = -12$

$d = -2$

Substitute into (3):

$2a_1 + 3(-2) = 16$

$2a_1 = 22$

$a_1 = 11$

$S_{10} = \left(\frac{10}{2}\right)(2(11) + 9(-2))$

$= 20$